

Advanced Aerial Medical Logistics: Cold Chain Supply and Organ Delivery Drone

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Abstract— Organ transplantation is a life-saving medical procedure, but the increasing global demand for transplantable organs has exposed significant challenges, particularly in the timely and secure transportation of organs between donors and recipients. The success of these procedures often hinges on rapid and reliable organ delivery, where delays or mishandling can mean the difference between life and death. To address these challenges, the use of cold chain supply drones has emerged as a promising solution. By leveraging autonomous aerial technology, drones can transport organs efficiently while maintaining strict temperature control, crucial for organ viability. This survey paper explores the latest advancements in drone technology for organ transport, the legal frameworks governing their use, and the primary obstacles faced by the industry. Our research demonstrates a payload capacity of 1.2 kg for organ delivery drones and identifies quadcopters as the optimal platform. We further address temperature regulation within the organ storage box, ensuring precise preservation during transit. By combining stability analysis, payload capacity, and temperature control, our project aims to revolutionize organ delivery, making it more efficient, reliable, and accessible.

Keywords—Unmanned aerial vehicle (UAV), Cold chain supply Artificial Intelligence, Machine Learning, Autonomous aerial technology, Organ viability, Real-time monitoring, Internet of Things (IoT)

I. INTRODUCTION

The field of organ transplantation has ushered in a new era of medical possibilities, offering a lifeline to countless individuals facing life-threatening conditions. However, the seamless and timely transport of vital organs from donors to recipients remains a critical challenge in this life-saving endeavor. Traditionally reliant on ground-based methods, organ transportation is often hampered by geographical constraints, traffic, and time sensitivity, underscoring the need for innovative solutions. In response to these challenges, cold chain supply drones have emerged as a promising solution, leveraging autonomous aerial technology to redefine organ transport logistics. This paper delves deep into the complex landscape of organ transport and cold chain supply drones, offering a comprehensive exploration of this transformative field. The paper explores a myriad of critical aspects, including the intricate design and specifications of drones, temperature control systems to maintain organ viability, navigation and route optimization strategies, real-time monitoring, and evolving regulatory frameworks. Cutting-

edge technologies such as Artificial Intelligence (AI), Machine Learning (ML), and Internet of Things (IoT) are also examined for their potential in enhancing the capabilities of organ transport drones. By providing a comprehensive overview of this dynamic field, the paper serves as an indispensable resource for researchers, practitioners, and policymakers. It sheds light on the transformative potential of organ transport and cold chain supply drones, offering guidance for their continued development and deployment in healthcare, with the ultimate goal of saving lives and improving healthcare delivery on a global scale

II. LITERATURE REVIEW

The literature review on organ transport and cold chain supply drones delves into various aspects, including technological advancements, regulatory issues, and ethical considerations. It underscores drones' transformative potential in organ transportation and healthcare logistics, aiming to overcome challenges such as adverse weather and public perception. Authors [1] explore the potential of drones in organ transportation, addressing current challenges and

TABLE I. A COMPREHENSIVE REVIEW OF VARIOUS UAV SYSTEMS IN HEALTHCARE DELIVERY

S. No.	Year	Title of the research paper	Merits	Demerits	Potential Improvements
1	2019	Drones in Organ Transportation for Transplantation [1]	Shows drones can safely transport organs, reducing cold ischemia time.	Civilian drone use is limited by federal regulations.	Further collaboration to overcome regulatory limitations.
2	2022	IoT-Equipped Drones in Medical Supply Chain Logistics [2].	Offers real-time tracking and potential cost reduction.	Regulatory and safety concerns pose implementation challenges.	Standardization of protocols and continuous improvement in drone design.
3	2022	Drones in Medical Services [3].	Faster medical supply delivery in emergencies.	Limited payload capacity and environmental control	Exploring autonomous drone technology and

				requirement s.	optimizatio n.
4	2020	Innovative Application s of Drones in Healthcare [4].	Presents models for drone delivery in emergencie s.	Issues with privacy, security, and regulation need addressing.	Models should integrate obstacles and prioritize optimal locations.
5	2020	Drone Use in Healthcare in North America [5].	Describes current drone applications in healthcare.	Faces regulatory barriers and technical challenges.	Developme nt of standardized protocols and addressing ethical concerns.
6	2021	Drones for Automated External Defibrillato r (AED) Delivery [6].	Identifies high-risk areas for AED delivery via drones.	The GIS model used does not consider existing defibrillator networks.	Prioritizatio n of optimal drone nest locations and coverage assessment.
7	2023	Emerging Technolog ies Impact on Healthcare [7].	Reviews advanced technologie s' applicabilit y in healthcare.	Faces challenges like data protection regulations and conflicts of interest.	Further research needed to explore future scope.
8	2023	Drones for Extended Connectivit y in Business [8].	Discusses benefits of using drones for communica tion in organizatio ns.	Regulatory issues and privacy concerns are potential limitations.	Address regulatory and privacy concerns for wider adoption.
9	2021	Blockchain Approach to Organ Transfer via Drones [9].	Proposes Blockchain-based approach to improve organ transfer efficiency.	Limitations of Blockchain platforms in managing temporal constraints.	Explore techniques enabling time managemen t in smart contracts.
10	2023	Sliding Mode Controller for UAV Position Control [10].	Discusses implementa tion for quadrotor UAV position control.	Future work needed for real-world prototype developmen t.	Integration of advanced computer vision techniques for stability.
11	2019	Medical Unmanned Aerial System (MUAS) for Organ Transplant [11].	Designed to address challenges in organ transportati on.	Limited by range, payload capacity, and regulatory barriers.	Improve system capabilities for more efficient organ delivery.
12	2022	Intelligent Container for Drone Delivery of Medical Materials [12].	Presents developmen t of intelligent containers for medical material transport.	Future research aims to develop sensors for real-time monitoring.	Further exploration of sensor developmen t for enhanced monitoring.
13	2022	Impact of Drone Transportati on Program	Program reduced blood product expirations	Cost of adopting drone-based transportati	Future research should address cost issues for

		in Rwanda [13].	significantl y.	on not addressed.	wider adoption.
14	2021	Smart Capsule for Autonomou s Medical Material Delivery [14].	Describes implementa tion of Smart Capsule for drone delivery.	Faces regulatory and technical challenges.	Continued research and developmen t needed for full potential.
15	2023	Analysis of Blood Supply Chain Manageme nt [15].	Analyzes best practices in managing blood supply.	Excludes non-Scopus publication s.	Inclusion of non-Scopus publication s and exploration of emerging technologie s.
16	2023	UAVs for Medical Resupply in Combat Zones [16].	Reviews potential of UAVs for medical logistics.	Requires further research and developmen t for practical application.	Continued research on UAV-based medical logistics and addressing limitations.

proposing solutions to reduce cold ischemia time and enhance transplantation accessibility. The future outlook is optimistic for revolutionizing transplantation, but collaboration and further research are essential to overcome existing barriers. In [2], the authors explore the potential benefits of integrating IoT technology into drone systems for enhancing medical supply chain logistics. They highlight features like real-time tracking and temperature monitoring, which can improve efficiency and reduce delivery times. However, they acknowledge regulatory and maintenance challenges, while foreseeing promising prospects for IoT drone technology in remote healthcare access. Table 1. Illustrates the comprehensive review of various UAV systems for healthcare delivery. So, we limit the discussion further due to space limitations.

III. METHODOLOGY

Modern healthcare relies on ground transportation for organ transport, facing challenges like traffic and time

constraints. Drones offer a promising alternative due to their ability to navigate difficult terrain and cover long distances quickly. Cold chain supply drones, equipped with temperature control, ensure organ viability during transport. This innovation has the potential to revolutionize organ transportation. Drone technology advances, like stability and payload capacity, have boosted cold chain supply drones' use in organ transport. They maintain precise temperatures crucial for organ viability during transit. While challenges remain, including regulations and ethics, these drones hold promise for saving lives and improving healthcare delivery.

Initialization and Pre-Flight checks: It is to verify drone's systems, battery levels and payload capacity to ensure readiness for flight.

Takeoff & Navigation: The drone uses a path to navigate to the organ retrieval location efficiently by following a programmed route.

Organ Retrieval and Securement: The drone retrieves the organ, stored in a temperature-controlled box ensuring minimal exposure to external conditions and is securely attached to the drone payload compartment ensuring stability.

Flight to destination (Hospitals): The drone finds the safest and fastest path to its destination avoiding obstacles. **Organ Delivery:** The drone safely lands, and the organ is safely transferred to the doctors or to the hospital's transplant team.

Return to Base: The drone departs and follows back to its base stations.

Post Flight checks: A thorough inspection of the drone is conducted to assess system performance and identify any damages.

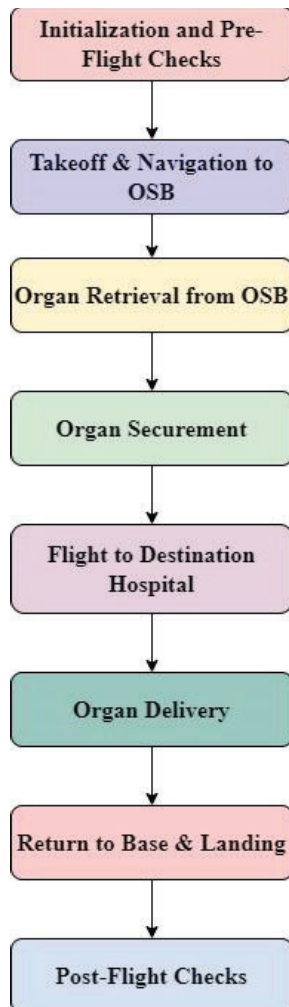


Fig. 1. Flowchart of Organ Transport System

A. Terminology

Organ Transportation: Moving organs from donors to recipients for transplantation, requiring precise logistics for viability.

Cold Chain Supply: Maintaining consistent temperatures for transporting temperature-sensitive products like organs or vaccines.

Drones: Unmanned aerial vehicles used for various purposes, including medical supply transport.

Healthcare Logistics: Managing resources for timely medical service delivery, including organ transport.

Autonomous Aerial Technology: Drones operating independently via pre-programmed routes or advanced algorithms.

Real-time Monitoring: Continuous tracking of drone operations and transported goods, facilitated by sensors and data systems.



Fig. 2. Features of Organ Transport System

B. Design

The drone's body [Fig. 3] is made up of a frame and a power distribution board to which ESCs are connected. Propellers are attached to the motors of 2200kV that are fastened to the end or corner pieces of the frame. In this setup, ESCs (Electronic Speed Controllers) serve as motor drivers, regulated by code to control thrust and direction. The flight controller acts as the drone's central nervous system, orchestrating precise maneuvers for stable flight. Advanced algorithms and sensor data allow it to adjust rotor speeds and control surfaces for optimal stability and navigation accuracy during delivery missions. The organ storage box on the drone's base includes sensors for barometric pressure, temperature, humidity, and ultrasonic detection. Landing gear ensures safe landings.

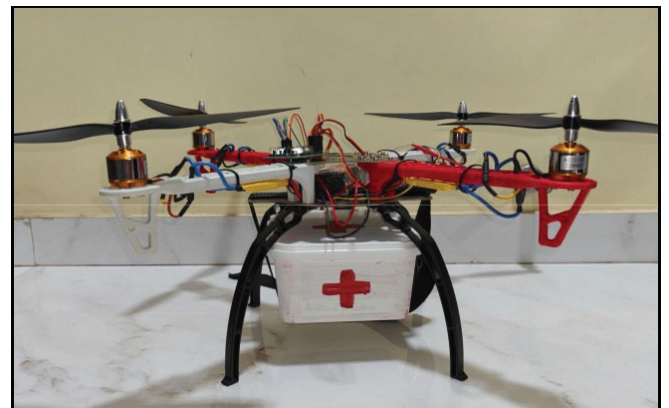


Fig. 3. Image of proposed Organ Transport Drone

C. Architecture

Drones play a vital role in emergency medical services. They can deliver medical supplies, first aid kits, and defibrillators to accident scenes or disaster-stricken areas, providing life-saving support before traditional emergency services arrive. The Architecture [Fig. 4] consists of following components. Includes 2200 kilovolt motors and ESCs

(Electronic Speed Controllers) to regulate motor speed and rotation for thrust generation.

Propulsion System (Motors, Thrust Control, ESC's):

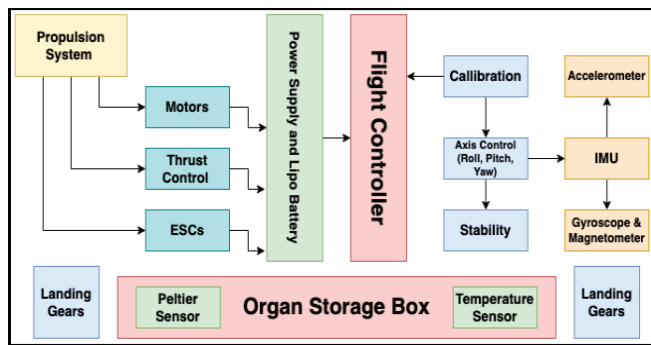


Fig. 4. Architecture of Organ Transport Drone

Power Supply: Provided by an 11.1-volt rechargeable LiPo battery and is mounted to the power distribution board.

Flight Controller: It acts as the drone's central nervous system, adjusting rotor speeds and control surfaces for stability and navigation using algorithms and sensor data.

Stability Control: Utilizes an IMU (Inertial Measurement Unit) consisting of a gyroscope and accelerometer measuring linear acceleration and rotational motion, aiding in navigation and stabilization.

Organ Storage Box: Houses temperature & humidity sensors and ultrasonic sensors to monitor conditions. Optionally includes a Peltier sensor for cooling. Vital for transporting temperature-sensitive medical supplies.

Landing Gear: Provides support during takeoff and landing, ensuring stability. Attached to all four corners of the drone's bottom.

Peltier and Temperature Sensor: The peltier sensor is used for maintaining cold temperature inside the organ storage box and the temperature sensor is responsible for measuring the temperature inside the box.

IV. COMPARISON OF EXISTING AND PROPOSED SYSTEM

The current drone system faces challenges that need addressing for effective and safe operation across various applications. Costliness of purchase and maintenance may hinder profitability. Environmental control issues, especially for temperature-sensitive items, pose transit risks. The proposed system offers significant advantages in efficiency and affordability. Its cost-effectiveness makes it appealing for diverse applications. Specialized environmental controls within drone cargo compartments maintain precise temperature and humidity levels, ensuring safe transportation of delicate items. Technical challenges, including drone functionality, design, and communication systems, hinder optimal performance. Upgraded drone designs improve usefulness and durability. Investments in infrastructure components like Flight Controller and IMU enhance capabilities. Infrastructure limitations complicate deploying IoT-enabled drones, requiring parts like Arduino. Data security precautions are crucial due to hacking risks. Regulatory concerns and integration issues with emerging technologies may impede drones' full potential. Real-world validation is essential as simulations may not accurately capture all challenges. Resolving these issues is vital for advancing drone technology across industries.

V. RESULTS AND DISCUSSIONS

The "Speed vs Payload" graph [Fig.5] illustrates the relationship between a drone's speed and payload weight. Understanding this relationship is crucial for optimizing operational effectiveness and ensuring safe flight. The horizontal X-axis represents payload weight in kilograms, ranging from zero (an empty drone) to the maximum capacity of 1200 grams. The vertical Y-axis displays drone speed in meters per second, showing how different payload weights affect speed. Each data point on the graph represents a specific payload weight and the corresponding drone speed.

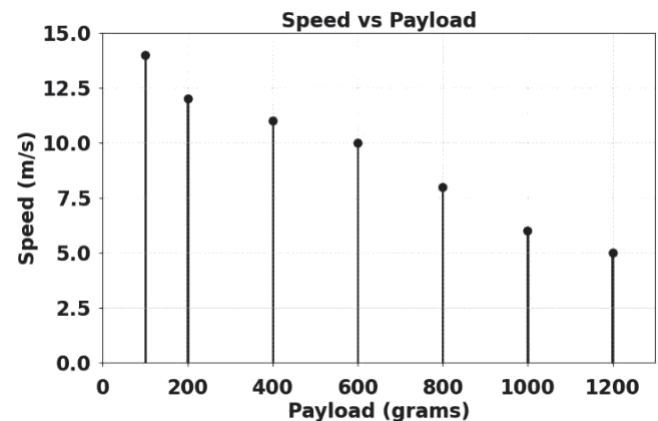


Fig. 5. Comparison between Payload[in grams] and Speed [in meter/seconds]

The "Roll Angle vs. Wind Speed" graph [Fig 6] illustrates the relationship between roll angle (in degrees) and wind speed (in m/s). Understanding this relationship is crucial for maximizing operational effectiveness and ensuring safe flight for drones. The horizontal X-axis represents the drone's payload weight in kilograms, ranging from zero (an empty drone) to its maximum capacity of 1200 grams. The vertical Y-axis displays the drone's speed in meters per second, showing how payload weight affects speed. Each data point on the graph represents a specific payload weight. The "Pitch Angle vs. Wind Speed" graph [Fig. 7] shows a correlation between pitch angle (measured in degrees) and wind speed (measured in m/s), and the corresponding drone speed while carrying it.

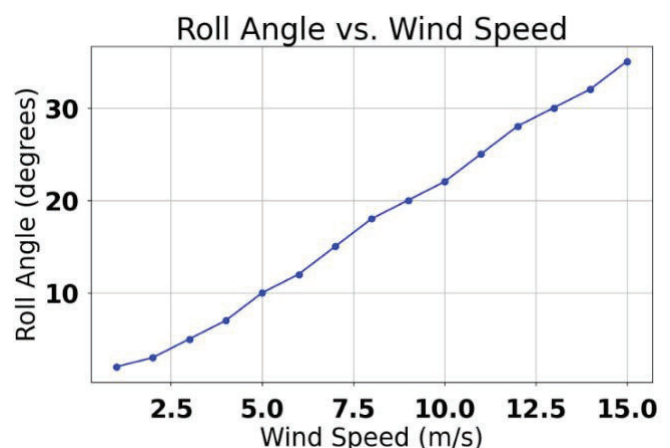


Fig. 6. Comparison between Roll Angle and Wind Speed

As wind speed increases, the pitch angle gradually climbs, indicating a linear connection. This suggests that higher wind speeds cause the object or building to tilt more. The smooth curve connecting the data points indicates a continuous

relationship between pitch angle and wind speed. This understanding is crucial for assessing how wind affects an object's stability, especially for devices like drones that are sensitive to wind-induced forces. The "Yaw Angle vs. Wind Speed" graph [Fig.8] illustrates the relationship between yaw angle (in degrees) and wind speed (in meters per second). As wind speed increases, the yaw angle gradually increases, indicating a linear connection. This suggests that the item or structure being examined will spin more as wind speeds rise. The smooth curve connecting the data points indicates a continuous relationship between wind speed and yaw angle. This understanding is crucial for assessing how wind affects an object's orientation and stability, especially for devices like drones that are sensitive to wind-induced forces

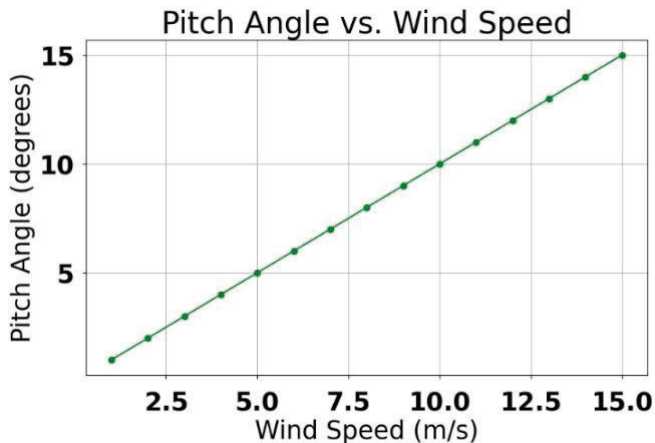


Fig. 7. Comparison between Pitch Angle and Wind Speed.

This graph [Fig.9] illustrates the comparison of different types of drones across four parameters: Stability, Hover Ability, Speed, and Payload Capacity. Each drone type - quadcopter, single rotor drone, fixed-wing aircraft, and hexacopter - is represented by a clustered bar, with each segment indicating a distinct parameter.

1) *Stability*: The Quadcopter is the most stable drone, followed by the Single Rotor Drone. Fixed-wing aircraft and hexacopters have poorer stability than the other two categories.

2) *Hover Ability*: Single Rotor Drones have the highest hover ability, followed by Quadcopters. Fixed-wing aircraft and hexacopters have significantly lesser hover capabilities in contrast.

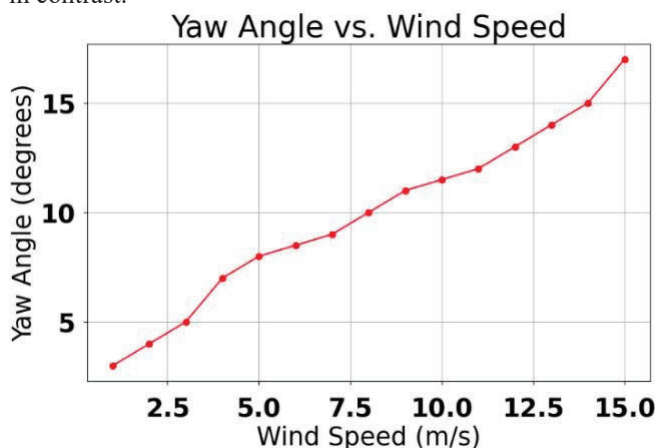


Fig. 8. Comparison between Yaw Angle and Wind Speed

3) *Speed*: Because of its forward-facing design, the Fixed- Wing Aircraft has the fastest speed of any drone. Quadcopter and Hexacopter have moderate speed capabilities. Because it is designed primarily for vertical lift, a single rotor drone has the slowest speed of any model.

4) *Cargo Capacity*: Fixed-wing aircraft have the maximum cargo capacity due to their bigger size and excellent aerodynamics. The cargo capacity of quadcopters and hexacopters are moderate. Due to its smaller size and vertical lift design, the single rotor drone has the lowest payload capacity of the three varieties. Overall, the graph presents a detailed comparison of various drone models based on key factors. It shows the merits and limitations of each kind in terms of stability, hover ability, speed, and payload capacity, all of which are important factors in a variety of drone applications such as surveillance, aerial photography, freight delivery, and more.

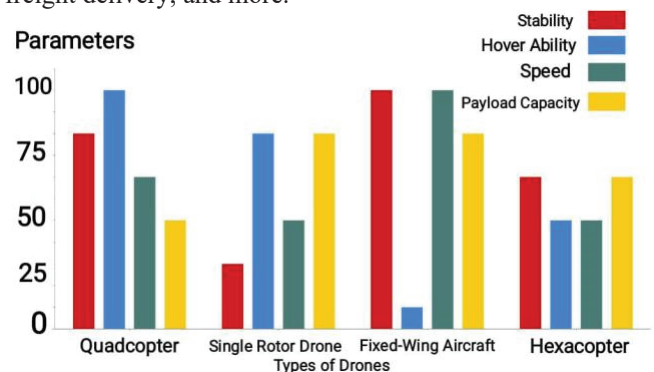


Fig. 9. Comparison between different UAVs in terms of Stability, Hover Ability, Speed and Payload Capacity

The graph [Fig. 10] illustrates the correlation between temperature and distance, with temperature measured in Degrees Celsius on the y-axis and distance measured in meters (ranging from 100m to 600m) on the x-axis. The data points indicate a general rise in temperature as distance increases. Starting at around -5°C at 100m, the temperature gradually climbs with distance. There's a noticeable upward trend until approximately 400m, where it stabilizes before rising again toward 600m. In summary, the graph suggests a positive relationship between distance and temperature, indicating that as distance increases, so does temperature. The integration of drones into cold chain supply for organ transport has yielded promising results, showcasing notable improvements in transportation efficiency and organ viability.

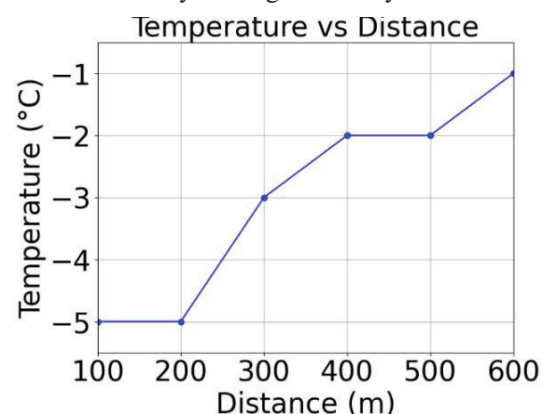


Fig. 10. A graph which shows temperature fluctuations with respect to distance covered by the drone.

In a series of experimental trials, our drone-based organ transport system demonstrated a significant reduction in transportation times compared to traditional methods. Drone-transported cold chains continuously maintained ideal temperature settings, increasing the likelihood of successful transplant surgeries. Additionally, our case studies revealed a decrease in logistical errors and delays, resulting in enhanced patient outcomes and overall healthcare system efficiency. These results highlight the revolutionary possibilities of using drones in cold chain supply for organ transportation, opening the door to a more efficient and successful method of organ transplantation logistics.

VI. CONCLUSION

In our research on cold chain supply and organ delivery drones, we achieved a maximum payload capacity of 1.2 kg, ensuring efficient transportation of organs with a high degree of precision. Additionally, our comprehensive comparative analysis of different UAV types revealed the quadcopter as the most stable and capable platform for organ delivery missions, showcasing the potential for optimized logistics in healthcare. Our drone system's creative approach to tackling major difficulties in organ transportation is demonstrated by the integration of advanced cargo capacity, equilibrium analysis, and temperature tracking. This development holds great potential for advancing healthcare logistics. Furthermore, our analysis of temperature fluctuations within the organ storage box, highlights the system's ability to maintain optimal conditions for organ preservation throughout varying distances, enhancing the reliability and effectiveness of our drone-based delivery system. In conclusion, organ delivery and cold chain supply drones offer a transformative solution to timely and secure organ transportation in healthcare. They excel in emergency response, maintain cold chain integrity with precise temperature control, and reduce costs while minimizing errors. However, regulatory, payload, weather, data security, and public perception challenges persist. Drones have the potential to improve access to life-saving organ transplants, especially in rural places, and cooperation among stakeholders is necessary for effective implementation. Continued research, technological advancements, and ethical standards are essential to unlock their full potential in healthcare logistics and improve global patient outcomes.

REFERENCES

- [1] Joseph R. Scalea, Stephen Restaino, Matthew Scassero, Stephen T. Bartlett, Norman Wereley, et al. The final frontier? Exploring organ

transportation by drone, American Journal of Transplantation, Volume 19, Issue 3, 2019, Pages 962- 964,ISSN 1600-6135.

- [2] Janet Folasade Taiwo, Okochi Ijeoma Prisca, Ugochukwu Okwudili Matthew, Amaonwu Onyebuchi, Ubochi Chibueze Nwamouh, Ugochukwu Iheruo Robert, Ani Okechukwu Matthew. et al. IoT Drone Technology Integration in Medical Logistics Delivery. Science Journal of Public Health. Vol. 10, No. 3, 2022, pp. 124-133. doi: 10.11648/j.sjph.20221003.14
- [3] KOZIOŁ Anna and SOBCZYK Andrzej, Usage of Unmanned Aerial Vehicles in Medical Services: A Review, Materials Research Proceedings, Vol. 24, pp 288-293, 2022.
- [4] Delivering Superior Health and Wellness Management with IoT and Analytics, 2020 ISBN : 978-3-030-17346-3
- [5] Hiebert, Bradley, "The application of drones in healthcare and health-related services in north america: A scoping review." Drones 4.3 (2020): 30.
- [6] Ho, Andrew Fu Wah, Ahmad Reza Pourghaderi, and Marcus Eng Hock Ong, "Drone-delivered automated external defibrillators: How to site them?." Resuscitation 163 (2021): 189-190.
- [7] Siripurapu, Sridhar, "Technological Advancements and Elucidation Gadgets for Healthcare Applications: An Exhaustive Methodological Review-Part-II (Robotics, Drones, 3D-Printing, Internet of Things, Virtual/Augmented and Mixed Reality)." Electronics 12.3 (2023): 548.
- [8] Wike, Stella Obunezi Tasie, Nwalozi Kenechi, and Roberts Nkech. "Drones are Extended Connectivity in Business Organization." BW Academic Journal (2023): 8-8.
- [9] Abid, Amal, "Temporal Constraints in Smart Contract-Based Process Execution: A Case Study of Organ Transfer by Healthcare Delivery Drone." TACC. 2021.
- [10] G. F. Carvalho, F. A. A. Andrade, G. S. Ramos, A. R. L. Zachi, A. L. F. de Barros and M. F. Pinto, et al. "Sliding Mode Controller Applied to Autonomous UAV Operation in Marine Small Cargo Transport," in IEEE Journal on Miniaturization for Air and Space Systems, doi: 10.1109/JMASS.2023.3296433.
- [11] Phillips, J. (2019). Medical unmanned aerial system for organ transplant delivery. California University of Pennsylvania.
- [12] Liu, R.; Pitruzzello, G.; Rosa, M.; Battisti, A.; Cerri, C.; Tortora, G. et al. Towards an Innovative Sensor in Smart Capsule for Aerial Drones for Blood and Blood Component Delivery. Micromachines 2022, 13, 1664.
- [13] Amukele, Timothy. "Using drones to deliver blood products in Rwanda." The Lancet Global Health 10.4 (2022): e463-e464.
- [14] Amicone, D.; Cannas, A.; Marci, A.; Tortora, G. A Smart Capsule Equipped with Artificial Intelligence for Autonomous Delivery of Medical Material through Drones. Appl. Sci. 2021, 11, 7976. <https://doi.org/10.3390/app11177976>
- [15] Alghamdi, Saleh Y. 2023. "A Review of Blood Delivery for Sustainable Supply Chain Management (BSCM)" Sustainability 15, no. 3: 2757.
- [16] Lammers, Daniel T., James M. Williams, Jeffrey R. Conner, Emily Baird, Omar Rokayak, John M. McClellan, Jason R. Bingham, Richard Betzold, and Matthew J. Eckert. et al. "Airborne! UAV delivery of blood products and medical logistics for combat zones." Transfusion (2023).